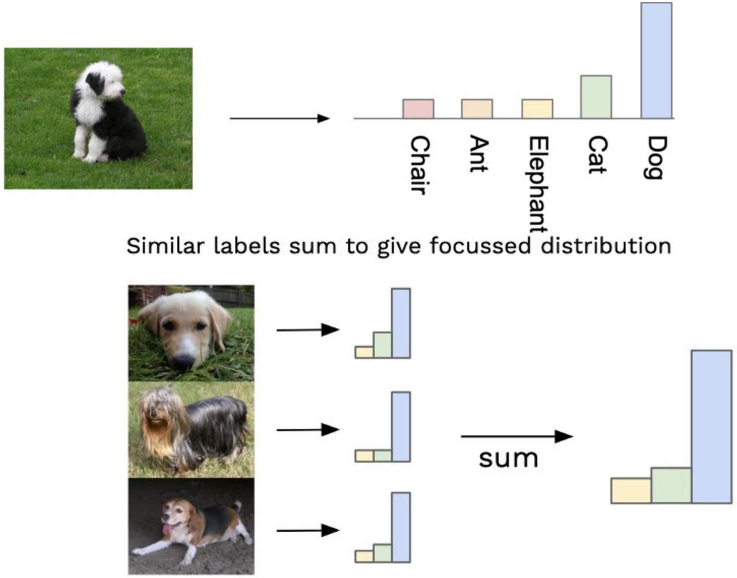
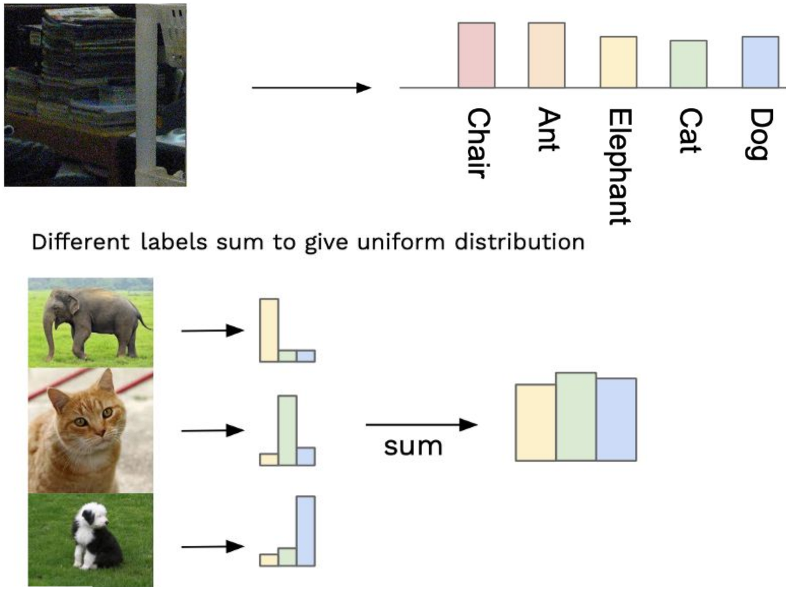
GANs Metrics Introduction

Lab Team

Inception Score (IS)

- Use InceptionNet with pretrained ImageNet
- Use to measure both quality and diversity





$$\mathrm{IS}(G) = \exp\left(\mathbb{E}_{\mathbf{x}\sim p_g} D_{KL}(\, p(y|\mathbf{x}) \,\|\, p(y) \,) \right)$$

=> The higher is the better

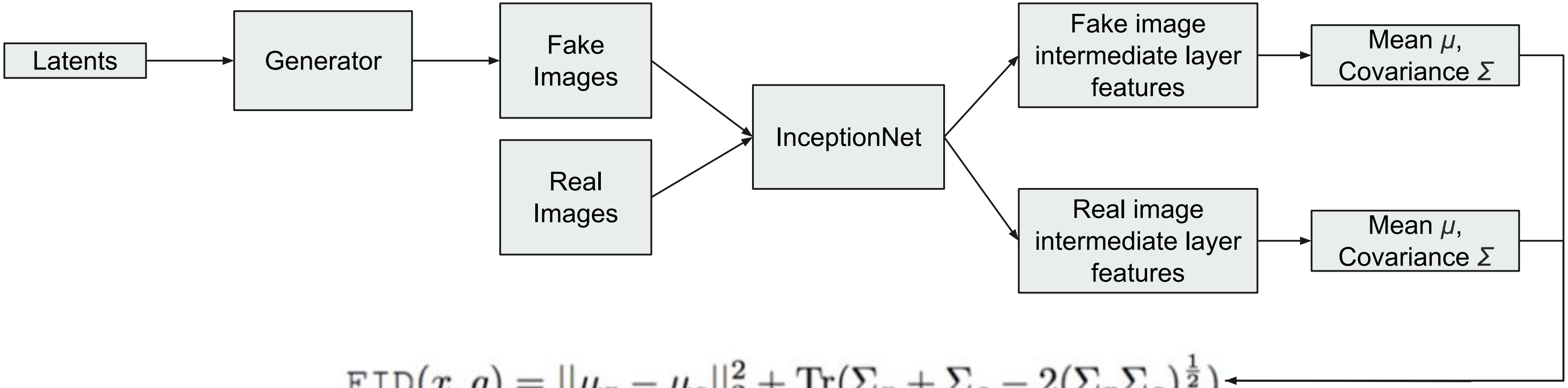
Inception Score (IS)

The cons of Inception Score

- The generator generates object that doesn't appear in the Inception pretrained
- The generator generates object whose label set is different from the Inception pretrained label set
- The feature of Inception pretrained (classification feature) doesn't represent the high-quality feature of images
- The generator generates only one good image each class => lack of diversity in each class but high IS
- The generator memorize and replicate real training data => doesn't create new content but high IS

Fréchet Inception Distance (FID)

- Use InceptionNet with pretrained ImageNet to extract features from intermediate layer
- Use to measure both quality and diversity
- The lower is the better



$$\mathrm{FID}(x,g) = ||\mu_x - \mu_g||_2^2 + \mathrm{Tr}(\Sigma_x + \Sigma_g - 2(\Sigma_x\Sigma_g)^{\frac{1}{2}})$$

Tr sums up all the diagonal elements

Conditional Inception Score (CIS)

- Use for evaluating multimodal image-to-image translation
- For each domain, train a classifier to classify image into classes of such domain (e.g.: domain "male" has some classes such as "long hair, short hair, bald, no hair")
- The higher is the better

$p(y_2|x_1) = \int p(y|x_{1\to 2})p(x_{1\to 2}|x_1)\, dx_{1\to 2}$ => should cover all classes in this domain
=> should have high entropy

$p(y_2|x_{1\to 2})$ => should belong to specific class in this domain
=> should have low entropy

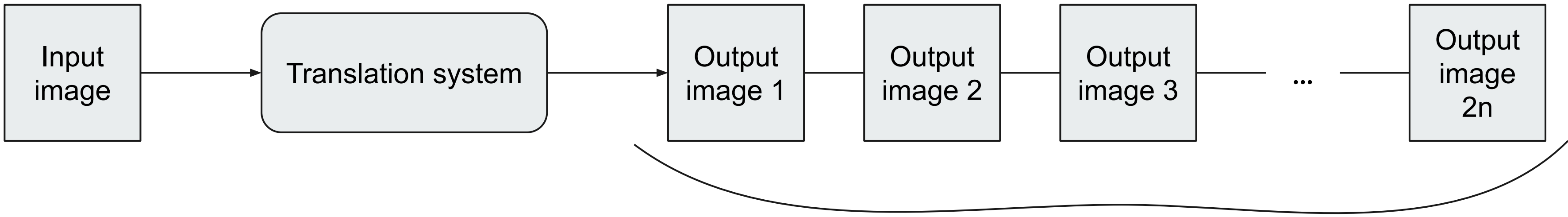
=> $\mathrm{CIS} = \mathbb{E}_{x_1\sim p(x_1)}[\mathbb{E}_{x_{1\to 2}\sim p(x_{2\to 1}|x_1)}[\mathrm{KL}(p(y_2|x_{1\to 2})||p(y_2|x_1))]]$

$p(y_2) = \iint p(y|x_{1\to 2})p(x_{1\to 2}|x_1)p(x_1)\, dx_1\, dx_{1\to 2}.$ => should all classes in this domain
=> should have high entropy

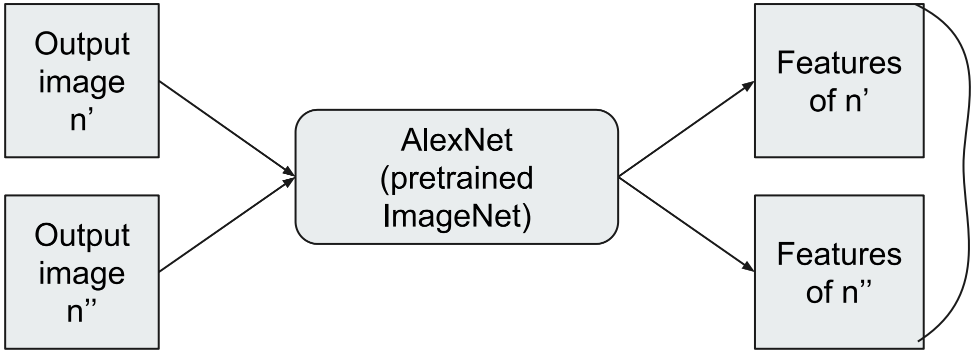
=> $\mathrm{IS} = \mathbb{E}_{x_1\sim p(x_1)}[\mathbb{E}_{x_{1\to 2}\sim p(x_{2\to 1}|x_1)}[\mathrm{KL}(p(y_2|x_{1\to 2})||p(y_2))]]$

LPIPS Distance

- Use to measure I2I translation diversity
- The higher is the better



Randomly split into n pairs



-> Calculate L2 distance of each pair
-> Calculate the weighted mean over n pairs
-> LPIPS distance

Perceptual Path Length (PPL)

- Use to “measure how drastic changes the image undergoes as we perform interpolation in the latent space”
- The lower is the better

$$l_{\mathcal{Z}} = \mathbb{E}\Big[\frac{1}{\epsilon^2} d\big(G(\mathrm{slerp}(\mathbf{z}_1, \mathbf{z}_2; t)), G(\mathrm{slerp}(\mathbf{z}_1, \mathbf{z}_2; t+\epsilon))\big)\Big]$$

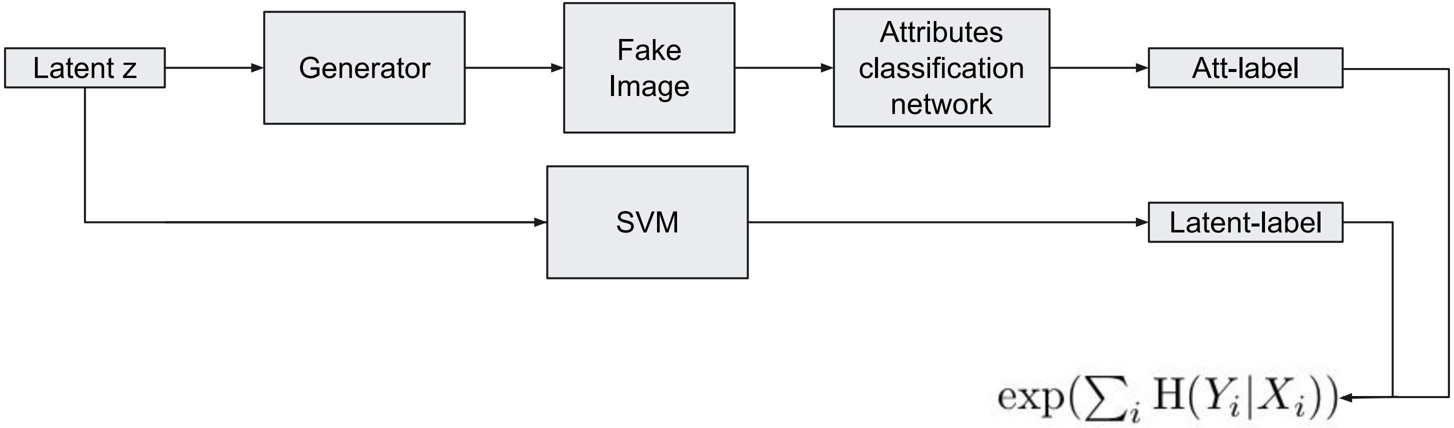
$\mathbf{z}_1, \mathbf{z}_2 \sim P(\mathbf{z})$
$t \sim U(0,1)$
slerp spherical interpolation
G is the generator
$d(\cdot,\cdot)$ evaluates the perceptual distance between the resulting images
$\mathbb{E}$ expectation by taking 100,000 samples
epsilon $\epsilon = 10^{-4}$

$$l_{\mathcal{W}} = \mathbb{E}\Big[\frac{1}{\epsilon^2} d\big(g(\mathrm{lerp}(f(\mathbf{z}_1), f(\mathbf{z}_2); t)), g(\mathrm{lerp}(f(\mathbf{z}_1), f(\mathbf{z}_2); t+\epsilon))\big)\Big]$$

lerp linear interpolation

Linear Separability

- Use to “measure how well the latent-space points can be separated into two distinct sets via a linear hyperplane”
- The lower is the better



X: classes predicted by the SVM
Y: classes determined by the pre-trained classifier
i: enumerates the 40 attributes in the CelebA dataset